

Problem Solving Using Python (CSE1012)

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Agenda / Contents



01 Numbers



02 Variables or Identifiers



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Data Types

- Variables can hold values of different types called *Data Types*
- Different data types needed to store different types of values in the variables
- Ex. A person's age is stored in number, Name in characters, address is a mixture of numbers and characters

Data Types

- Python has various standard data types that are used to define operations possible on them and the storage method for each of them
- We can even create our own data types in Python (like classes)

Data Types

- The five standard data types supported by Python includes:
 - Numbers
 - String
 - List
 - Tuple
 - Dictionary

Assigning or Initializing Variables

- In Python, you need not explicitly declare variables to reserve memory space
- Declaration is done automatically when a value is assigned to the variable using equal sign (=)
- Operand on the left side of (=) is the name and right side is the value to be stored

Assigning or Initializing Variables

Assigning Values to Variables

num = 7

amt = 123.45

code = 'A'

msg = "Hello"

print("The Number is: ",num)

print("The Character is: ",code)

print("The Message is: ",msg)

Re-assigning Values to Variables

- In Python, you can re-assign values to variables as many times as needed
- You may change data types of the same variable in subsequent statement

Re-assigning Values to Variables

#Program to re-assign values to Variables

Val = 2

print(Val)

Val = "Hello"

print(Val)

Val = 23.5

print(Val)

Multiple assignment

- Python allows you to assign a single value to multiple variables simultaneously

- Ex.

```
sum = flag = a = b = 0
```

Multiple assignment diff data type

- Python also allows assigning multiple variables with different data types

- Ex.

`sum, a, b, msg = 0, 2, 3, "RESULT"`

Multiple statement in single line

- Multiple statements in a single line should be separated by a semicolon

- Ex.

```
Msg = "Hello"; print(Msg)
```

Boolean Data Type

- **Boolean is another data type in Python.**
- **A variable of Boolean type can have one of the two values- True or False.**
- **Similar to other variables, the Boolean variables are also created while we assign a value to them or when we use a relational operator on them.**

Boolean Data Type

<pre>>>> Boolean_var = True >>> print(Boolean_var) True</pre>	<pre>>>> 20 == 30 False</pre>	<pre>>>> "Python" == "Python" True</pre>
<pre>>>> 20 != 20 False</pre>	<pre>>>> "Python"! = "Python3.4" True</pre>	<pre>>>> 30 > 50 False</pre>
<pre>>>> 90 <= 90 True</pre>	<pre>>>> 87 == 87.0 False</pre>	<pre>>>> 87 > 87.0 False</pre>
<pre>>>> 87 < 87.0 False</pre>	<pre>>>> 87 >= 87.0 True</pre>	<pre>>>> 87 <= 87.0 True</pre>

Programming Tip: <, > operators can also be used to compare strings lexicographically.

User Input

- Python uses `input()` function to allow user input
- This function takes user input as string

User Input

```
name = input("What's your name?")  
age = input("Enter your age : ")  
print(name + ", you are " + age + " years old")
```

OUTPUT

```
What's your name? Goransh  
Enter your age : 10  
Goransh, you are 10 years old
```

Comments

- **Comments** are the non-executable statements in a program.
- They are just added to describe the statements in the program code.
- It makes the program more readable
- The interpreter simply ignores the comments.

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Comments

- In Python, a hash sign (#) that is not inside a string literal begins a comment.
- All characters following the # and up to the end of the line are part of the comment

Indentation

- Whitespace at the beginning of the line is called *indentation*.
- These *whitespaces* or the *indentation* are very important in Python.

Indentation

- **In a Python program, the leading whitespace including spaces and tabs at the beginning of the logical line determines the indentation level of that logical line.**

Indentation

```
age = 21
    print("You can vote") # Error! Tab at the start of the line
Traceback (most recent call last):
  File "C:\Python34\Try.py", line 2
    print("You can vote")
    ^
IndentationError: unexpected indent
```

Reserved Words

- Reserved or keywords have pre-defined meaning
- They are not available for naming variables/Identifiers

Reserved Words

The following identifiers are used as reserved words, or *keywords* of the language, and cannot be used as ordinary identifiers. They must be spelled exactly as written here:

False	await	else	import	pass
None	break	except	in	raise
True	class	finally	is	return
and	continue	for	lambda	try
as	def	from	nonlocal	while
assert	del	global	not	with
async	elif	if	or	yield

Thank You

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